CSE6242 Spring2026 Project Proposal

Change-Point-Based Detection and Visual Analysis of Baseball Performance Shifts

Team26: Suho Lee, Hsiang Wen Hsiao, I Lin Tsai, Qixiang Goh, Xueying Jin

1. INTRODUCTION (Q1, Q4)

Identifying performance shifts before they manifest in season-long averages is the ultimate competitive advantage. We are developing an automated diagnostic dashboard by utilizing a Change-Point Detection (CPD) framework to pinpoint the exact onset of player slumps or breakouts. While current visualization tools provide vast amounts of player data, they often leave coaches buried in "noise." Our solution addresses this context gap by automating the detection of performance inflection points and visually diagnosing the root cause—distinguishing between mechanical changes, such as Exit Velocity or Launch Angle shifts, and psychological "clutch" performance dips. By providing a streamlined visual contrast of a player's profile before and after a shift, we offer high-speed diagnostic intelligence that transforms raw Statcast data into a decisive "first-mover" advantage for MLB front offices and fantasy enthusiasts alike.

2. METHODS (Q2)

Methods - Variables Selection

Given that baseball is a domain-intensive field, our methodology starts with careful variable selection. Taylor (2017) demonstrates exit velocity and launch angle possess significant predictive power under linear regression frameworks. Albert and Glickman (2020) further highlight the need to transform metrics depending on modeling objectives. Albert (2022) synthesizes the evolution of baseball metrics across offense, defense, and pitching, consolidating key performance indicators from traditional sabermetrics to modern Statcast variables. Together, these works justify beginning our variable selection from the three core domains of baseball—offense, defense, and pitching. However, these studies treat variables descriptively rather than examining their dynamic interactions over time.

Methods - Algorithms

Our framework follows the unified structure of change point detection (CPD) articulated by Truong, Oudre, and Vayatis (2020), which formalizes structural analysis through three components: Cost Function (what changes?) + Search Method (how to find?) + Constraint (how many changes?). Killick, Fearnhead, and Eckley (2012) detect mean or distributional shifts using the PELT algorithm. PELT applies dynamic programming to locate breakpoints and uses a penalized likelihood to control the number of detected changes. Whalen, Matthews, and Mills (2025) focus on multivariate mean shifts in baseball time series. They employ Double CUSUM statistics combined with Sparsified Binary Segmentation to locate structural breaks across multiple metrics simultaneously. Adams and MacKay (2007) detect distributional shifts using a Bayesian online framework, updating posterior run-length probabilities with prior-controlled change frequency. He and Wu (2019) propose the Parallel-Sum algorithm to detect mean-level structural shifts efficiently in large-scale data. Glazer (2025) similarly prioritizes tractable implementations, ensuring computational feasibility in applied sports analytics. Beyond the CPD framework, we also survey complementary statistical modeling approaches relevant to signal detection. Jensen, McShane, and Wyner (2009) stabilize noisy performance through hierarchical Bayesian shrinkage, refining the signal for detection. Bailey (2020) defines structural change in terms of shifts in outcome probability under a binomial likelihood framework. Hamilton (1989) conceptualizes nonstationarity as transitions between latent performance regimes rather than isolated breakpoints. In synthesis, we adopt Truong et al. (2020) as the computational framework while leveraging the detailed detection methods proposed in the subsequent literature. However, these studies largely lack a visualization methodology; therefore, we extend the framework with visualization techniques to enhance interpretability.

Methods – Visualization

Current research shows a move toward high-interactivity, yet significant gaps remain in diagnostic context. Lage et al. (2016) developed the StatCast Dashboard to query and filter spatiotemporal MLB data; however, their process often leads to visual noise and cognitive overload as data complexity scales across multiple games. Similarly, while Ono, Freire, and Silva (2021) demonstrate the power of interactive visualizations within computational environments like Jupyter Notebooks, these implementations are frequently hampered by performance latency and a lack of narrative guidance for the end-user. Together, these works high-

light a need for visualization processes that balance technical interactivity with better scalability and human-centered storytelling to bridge the "context gap" in sports analytics.

3. IMPLEMENTATION AND INNOVATION (Q3, Q5, Q6)

Computation Innovation

Building upon the CPD methodologies reviewed above, our computational contribution lies in adapting structural detection to high-frequency, multivariate Statcast data. Rather than applying generic change-point algorithms directly, we aim to examine coordinated shifts across mechanical indicators such as exit velocity, launch angle, and barrel rate, assessing cross-metric consistency when structural changes occur. This represents an application-level innovation tailored to domain-specific sports tracking data. Instead of focusing solely on isolated breakpoints in a single variable, we plan to interpret detected shifts through the joint behavior of multiple performance dimensions. By segmenting the time series into regime-based intervals and quantifying standardized differences, we seek to distinguish sustained structural adjustment from short-term statistical fluctuation. This reframes change detection from a univariate signal into a domain-oriented diagnostic framework grounded in multivariate evidence.

Visualization Innovation

Our innovation for visualization lies in the transition from static longitudinal tracking to automated comparative diagnostics through a specialized "Change-Point" visualization layer. Unlike traditional dashboards that require manual scanning of time-series data, our system proactively visualizes the precise "inflection point" of performance shifts. By utilizing a dual-state "Snapshot" UI, we provide an immediate visual contrast of mechanical metrics, such as Exit Velocity and Launch Angle, before and after a shift occurs. This removes the "context gap" identified in previous research by directly linking statistical fluctuations to physical behavioral changes. Furthermore, we provide a "Clutch vs. Core" performance breakdown that transforms Statcast data into an actionable diagnostic view.

Integrated Innovation and Comparative Advantage

Many traditional baseball metrics rely on season-long averages or percentages. While useful, these measures act as lagging indicators that often conceal meaningful performance shifts until a player has already experienced an extended decline. This project addresses these limitations by combining high-fidelity Statcast bat-tracking metrics with Change-Point Detection (CPD) algorithms to reduce the time required for coaches to identify performance inflection points. Our system supports real-time detection through algorithms such as Bayesian online inference, which estimates the run length of performance streaks as new data arrives, or the Pruned Exact Linear Time (PELT) algorithm, which scans historical data efficiently to identify structural breakpoints. If successful, this system could shift baseball coaching from a reactive process based on lagging indicators to a proactive diagnostic framework, enabling earlier identification of performance changes and distinguishing structural decline from temporary variance. Success will be measured through detection precision and recall against documented historical events like if a player rank increased or decreased after a change point detection is recorded. The risk is setting the constraints wrongly leading to over-detecting noise (flagging "bad luck" as a physical change).

4. PLAN OF ACTIVITIES (Q7, Q8, Q9)

Cost: $0. The project uses only open-source resources, including Statcast data and Python libraries (ruptures, D3.js, Plotly Dash) for computation and visualization. No cloud services or paid APIs are required.

Timeline: The project follows three main phases across February to April 2026. Phase I (2/16–3/5): topic selection and literature review. Phase II (3/3–3/30): data preparation and CPD development. Phase III (3/31–4/25): evaluation and final dashboard.

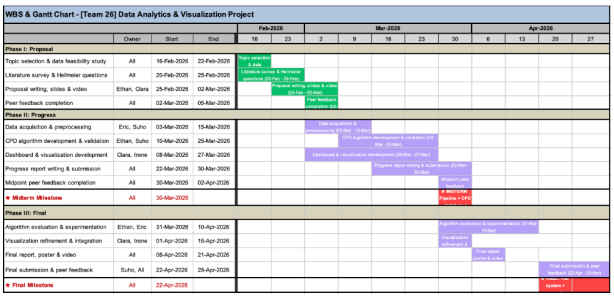

Figure 1. Project timeline (Gantt chart).

Midterm milestone (3/30): Deliverables include (1) a cleaned Statcast dataset, (2) a working CPD prototype for detecting performance change points, and (3) a prototype dashboard presenting player-level results.

Final milestone (4/22): Evaluation includes (1) measuring detection precision and recall against documented events, (2) comparing CPD algorithms (CUSUM, PELT, Bayesian), and (3) completing the final four-view dashboard and project deliverable.

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